

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

0

Question Number : 153 **Question Id :** 6406531133838 **Question Type :** SA
Correct Marks : 4

Question Label : Short Answer Question

Suppose the score is calculated as follows:
$$\text{score} = \# \text{matches} - 2 \times \# \text{mismatches} + \text{total gap penalty}$$

For every gap of length k , the gap penalty is calculated as $\sigma - (k - 1)$, where the gap opening penalty σ is computed from the score matrix S given in Table 2. The total gap penalty is the sum of the gap penalties for every gap.
What is the difference in the alignment score?
Enter the value as a single integer.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Set
Text Areas : PlainText
Possible Answers :

3

-3

Industry 4.0

Section Id :	64065380103
Section Number :	7
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	11
Number of Questions to be attempted :	11
Section Marks :	20
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No

Enable Mark as Answered Mark for Review and Clear Response : No
Section Maximum Duration : 0
Section Minimum Duration : 0
Section Time In : Minutes
Maximum Instruction Time : 0
Sub-Section Number : 1
Sub-Section Id : 640653172874
Question Shuffling Allowed : No

Question Number : 154 Question Id : 6406531133839 Question Type : MCQ
Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

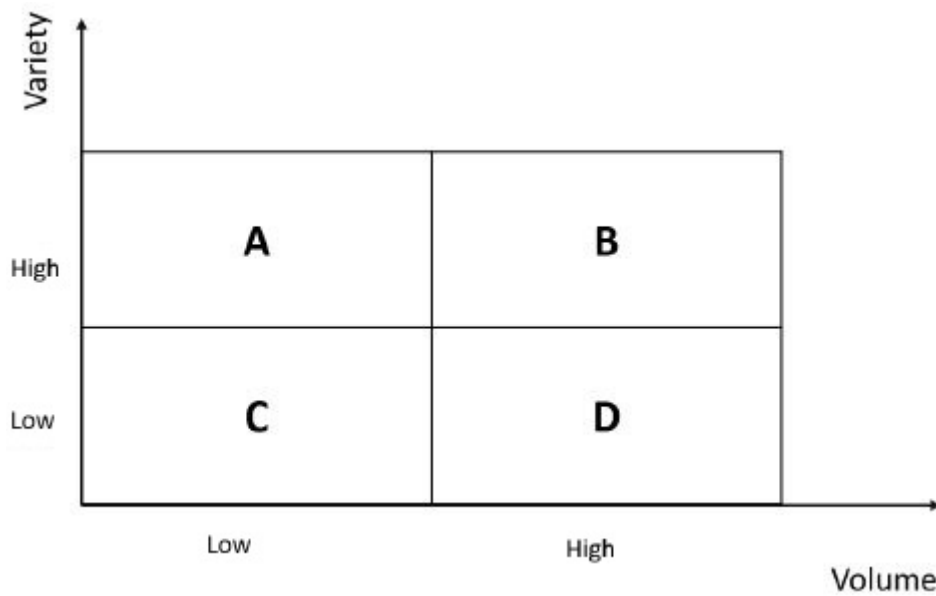
Options :
6406533830643. ✓ YES
6406533830644. ✗ NO

Sub-Section Number : 2
Sub-Section Id : 640653172875
Question Shuffling Allowed : Yes

Question Number : 155 Question Id : 6406531133840 Question Type : MCQ
Correct Marks : 1

Question Label : Multiple Choice Question

Select the appropriate layout for Quadrants A, B, D in the following figure:



Options :

6406533830645. ✖ A: Group layout, B: Product layout , D: Process layout

6406533830646. ✖ A: Product layout, B: Process layout , D: Group layout

6406533830647. ✔ A: Process layout, B: Group layout , D: Product layout

Question Number : 156 Question Id : 6406531133841 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following is an advantage of Process Layout?

Options :

6406533830648. ✖ Low material handling

6406533830649. ✖ Total production time per unit is small

6406533830650. ✖ Low training cost for the operators

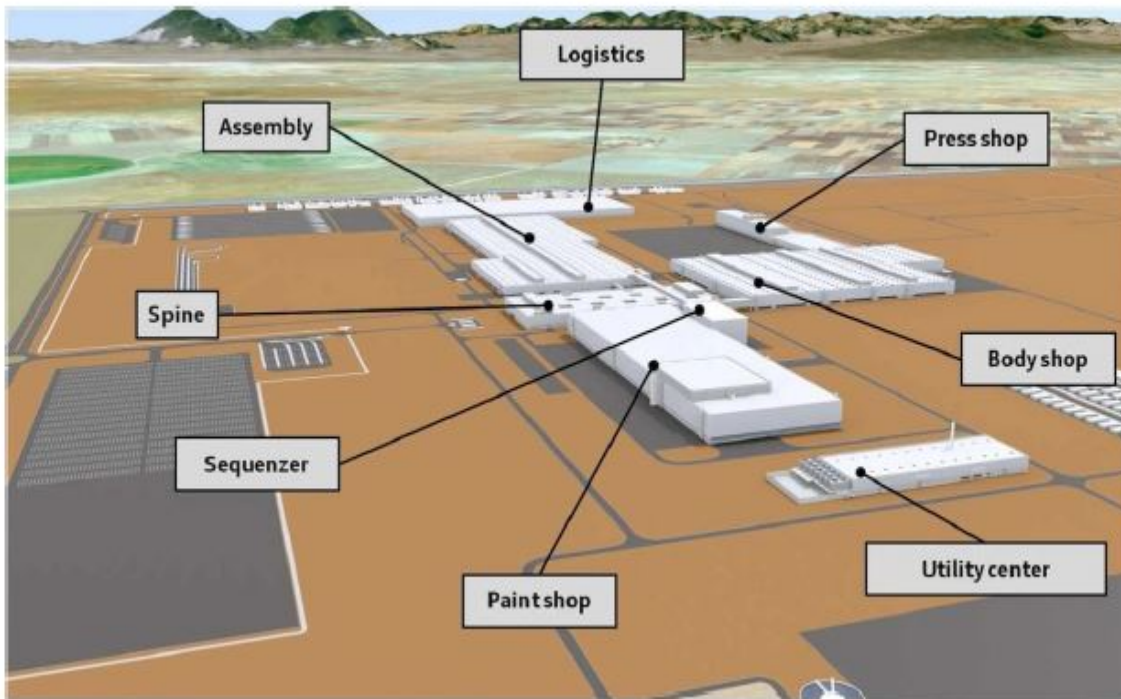
6406533830651. ✔ It can accommodate high variety of products

Question Number : 157 Question Id : 6406531133842 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Following is the layout from an aircraft manufacturing company. Identify the layout from the figure below.



Options :

- 6406533830652. ✖ Product layout
- 6406533830653. ✖ Process Layout
- 6406533830654. ✖ Group layout
- 6406533830655. ✔ Fixed position layout.

Sub-Section Number : 3
 Sub-Section Id : 640653172876
 Question Shuffling Allowed : Yes

Question Number : 158 Question Id : 6406531133843 Question Type : MCQ
 Correct Marks : 1.5
 Question Label : Multiple Choice Question

Match Col A with Col B for landmark outcome of industrial era

Col A (Industrial Era)	Col B (Major Outcome)
1.0	Digitization
2.0	CNC
3.0	Steam Engine
4.0	Mass production

Options :

- 6406533830656. ✖ 1.0:CNC, 2.0:Mass production, 3.0: Digitization, 4.0:Steam Engine
- 6406533830657. ✖ 1.0:Steam Engine, 2.0:mass production, 3.0: Digitization, 4.0:CNC
- 6406533830658. ✖ 1.0:Mass production, 2.0:Steam Engine, 3.0: Digitization, 4.0:CNC
- 6406533830659. ✔ 1.0:Steam Engine, 2.0:mass production, 3.0: CNC, 4.0: Digitization

Sub-Section Number : 4
 Sub-Section Id : 640653172877
 Question Shuffling Allowed : Yes

Question Number : 159 Question Id : 6406531133844 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

When will of the following scenarios, a facility layout problem is relevant?

Options :

6406533830660. ✓ A new facility is added

6406533830661. ✓ A new product is introduced

6406533830662. ✓ A new service is introduced

6406533830663. ✓ Shift in technology

Question Number : 160 Question Id : 6406531133845 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Identify the examples for Offshoring from the list below:

Options :

6406533830664. ✓ Shell, an oil company based in Netherlands, setting up a data science team in India.

6406533830665. ✗ Western Australia's electricity network agency assigns Tata Consultancy Services at Pune, to support it with an overhaul of its legacy IT systems, aimed at delivering cheaper and more reliable power.

6406533830666. ✓ Target Corporation, a retail giant based in US, setting up Supply Chain Optimization team at Bengaluru.

6406533830667. ✗ Tata Motors assigns the manufacturing of its braking system to TACO group.

Question Number : 161 Question Id : 6406531133846 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the correct statements from below:

Options :

6406533830668. ✗ Optimal solution in a facility location problem using Euclidean method will result in a region

6406533830669. ✓ Optimal solution in a facility location problem using Euclidean method rarely matches the center of gravity

6406533830670. ✓ Optimal solution in a facility location problem using Euclidean method will result in a point

6406533830671. ✗ Optimal solution in a facility location problem using Euclidean method matches the center of gravity in not of the cases

Sub-Section Number :

5

Sub-Section Id :

640653172878

Question Shuffling Allowed :

Yes

Question Number : 162 Question Id : 6406531133847 Question Type : MSQ

Correct Marks : 1.5 Max. Selectable Options : 0

Question Label : Multiple Select Question

There are four demand points at A(1,2), B(2,3), C(3,5), D(4,1). The respective demand at these points are 7, 1, 3, 5. Select the optimal location based on cross-median approach.

Options :

6406533830672. ✓ (2,2)

6406533830673. ✓ (3,2)

6406533830674. ✓ (2.5,2)

6406533830675. ✗ (2,3)

Sub-Section Number :

6

Sub-Section Id :

640653172879

Question Shuffling Allowed :

No

Question Id : 6406531133848 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (163 to 164)

Question Label : Comprehension

A factory works for two 8-hour shifts each day and produces a total of 30,000 electric bulbs using a set of workstations. There are 8 activities required to manufacture the bulb, such that the sum of all activity times is equal to 12 seconds. Answer the given subquestions:

Sub questions

Question Number : 163 Question Id : 6406531133849 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the cycle time (in seconds) at any workstations (round your answer to one decimal).

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.7 to 2.0

Question Number : 164 Question Id : 6406531133850 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the theoretical minimum number of workstations required?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

7

Sub-Section Number : 7

Sub-Section Id : 640653172880

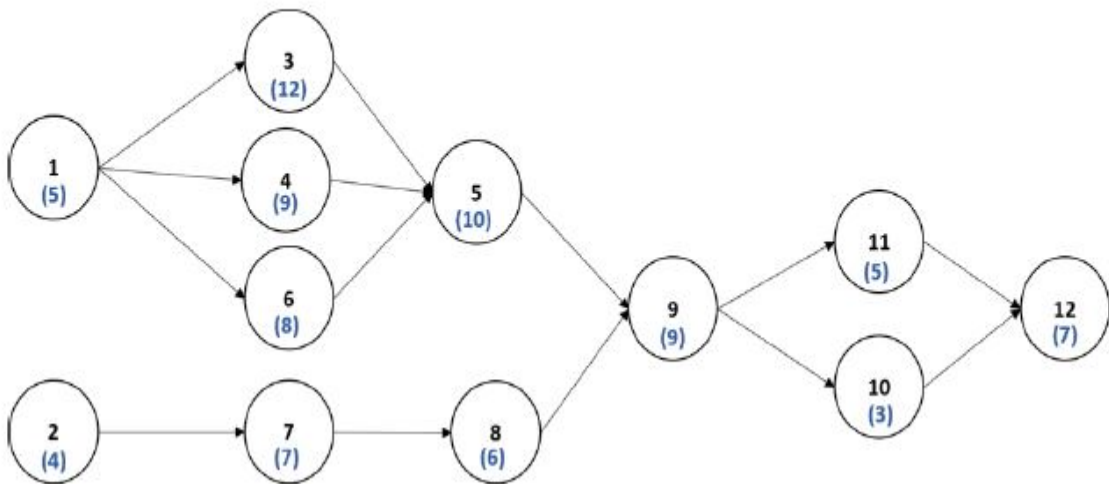
Question Shuffling Allowed : No

Question Id : 6406531133851 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (165 to 169)

Question Label : Comprehension

Consider the following precedence diagram. It has 12 activities. The numbers in blue and withing the braces denote the respective activity times in minute.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 165 Question Id : 6406531133852 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the minimum cycle time of the system? (in min)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12

Question Number : 166 Question Id : 6406531133853 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the maximum cycle time of the system? (in min)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

85

Question Number : 167 Question Id : 6406531133854 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

What is the balance delay of the following layout (round it to two decimals)

(1,2) -> (3) -> (4) -> (5) -> (6,7) -> (8,9) -> (10,11,12)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

18.5 to 19.5

Question Number : 168 Question Id : 6406531133855 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the efficiency of the layout, Mentioned in the previous question? (Round off to two decimals without percentage sign)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

80 to 81

Question Number : 169 Question Id : 6406531133856 Question Type : MCQ

Correct Marks : 1